Keyboard

File: BIT90_Keyboard.kicad_sch

Power

File: BIT90_Power.kicad_sch

Processor

File: BIT90_Processor.kicad_sch

Video Control

File: BIT90_Video_Control.kicad_sch

Joystick, Audio, Tape Out

File: BIT90_Joystick_Audio_TapeOut.kicad_sch

Video Encoder (PAL)

File: BIT90_Video_Encoder_PAL.kicad_sch

>> WORK IN PROGRESS <<

Open issues:

[1] Audio clock - where from? It looks to be labelled as AUDΦ (expansion pin 35 & SN76489 clock) but can't see it being generated?

[2] Floating input on CTRL1 buffer, output connected to D7 ... ?

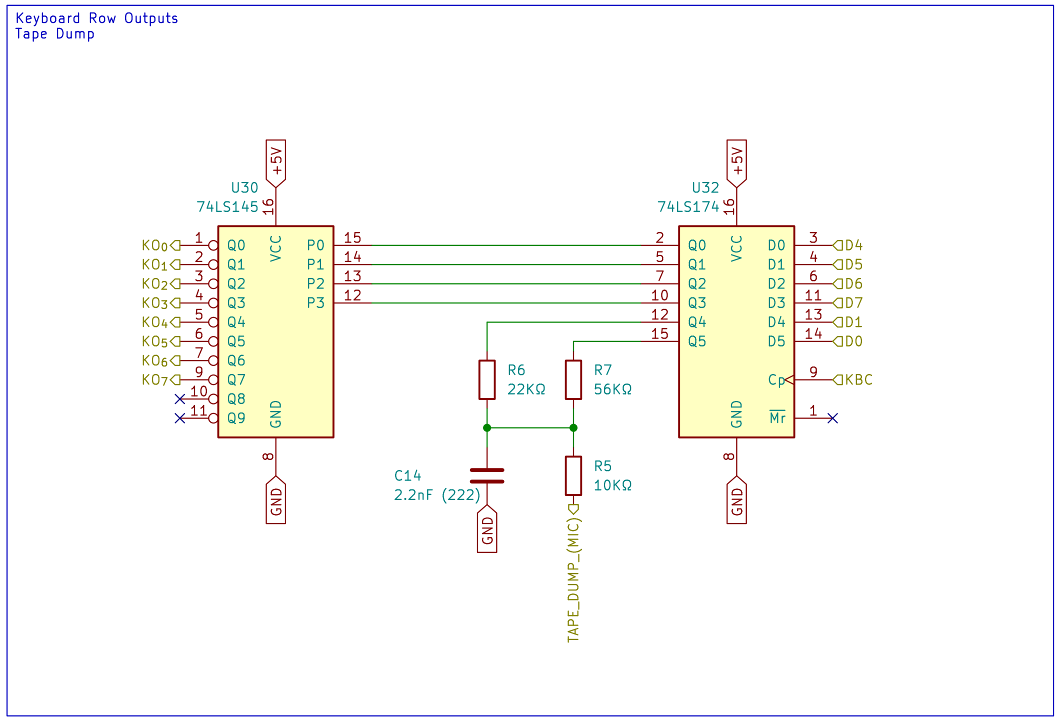
Interfaces

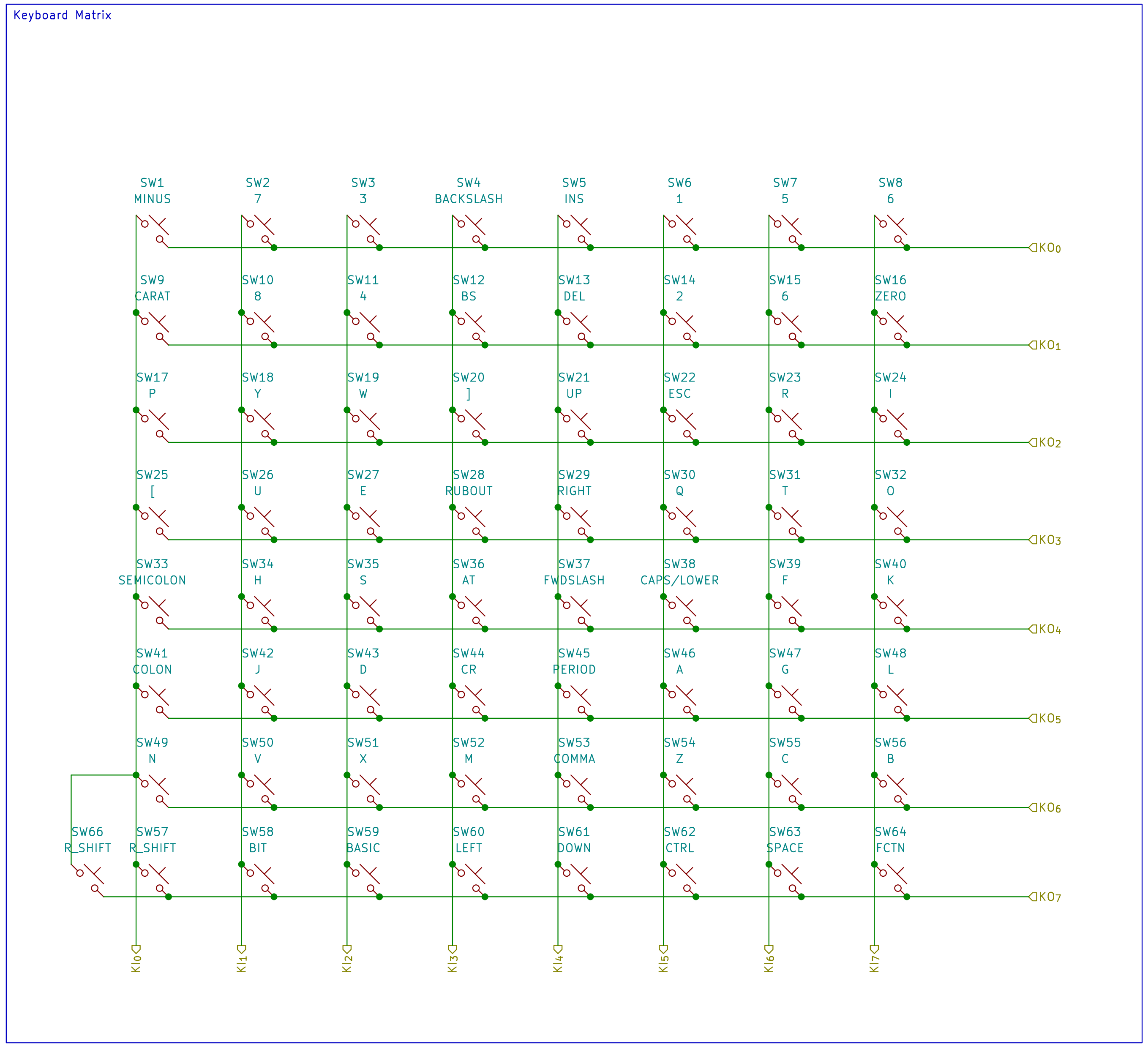
File: BIT90_Interfaces.kicad_sch

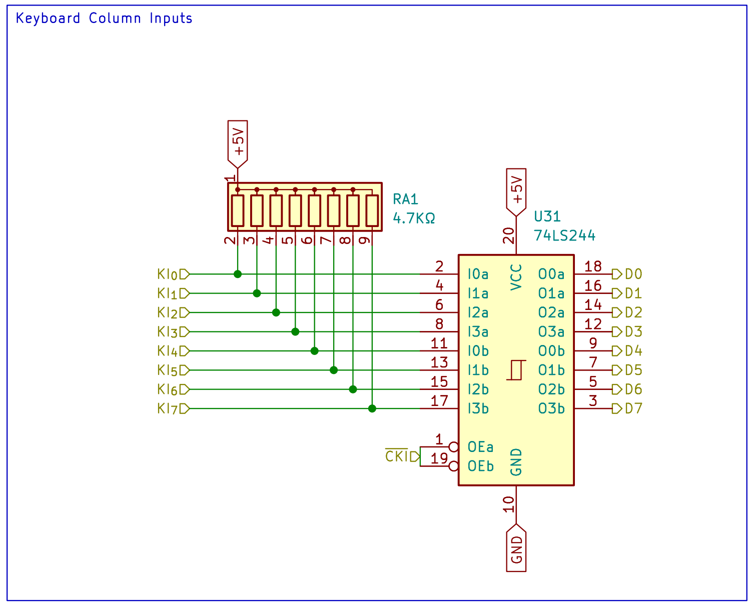
Spare Gates

File: BIT90_Spares.kicad_sch

Based on original schematics from BIT Corporation (普澤有限公司)		
Brett Hallen		
Sheet: /		
File: BIT90_Original.kicad_sch		
Title: BIT90 (PAL) Computer		
Size: A3	Date: 5/JUN/2026	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/9







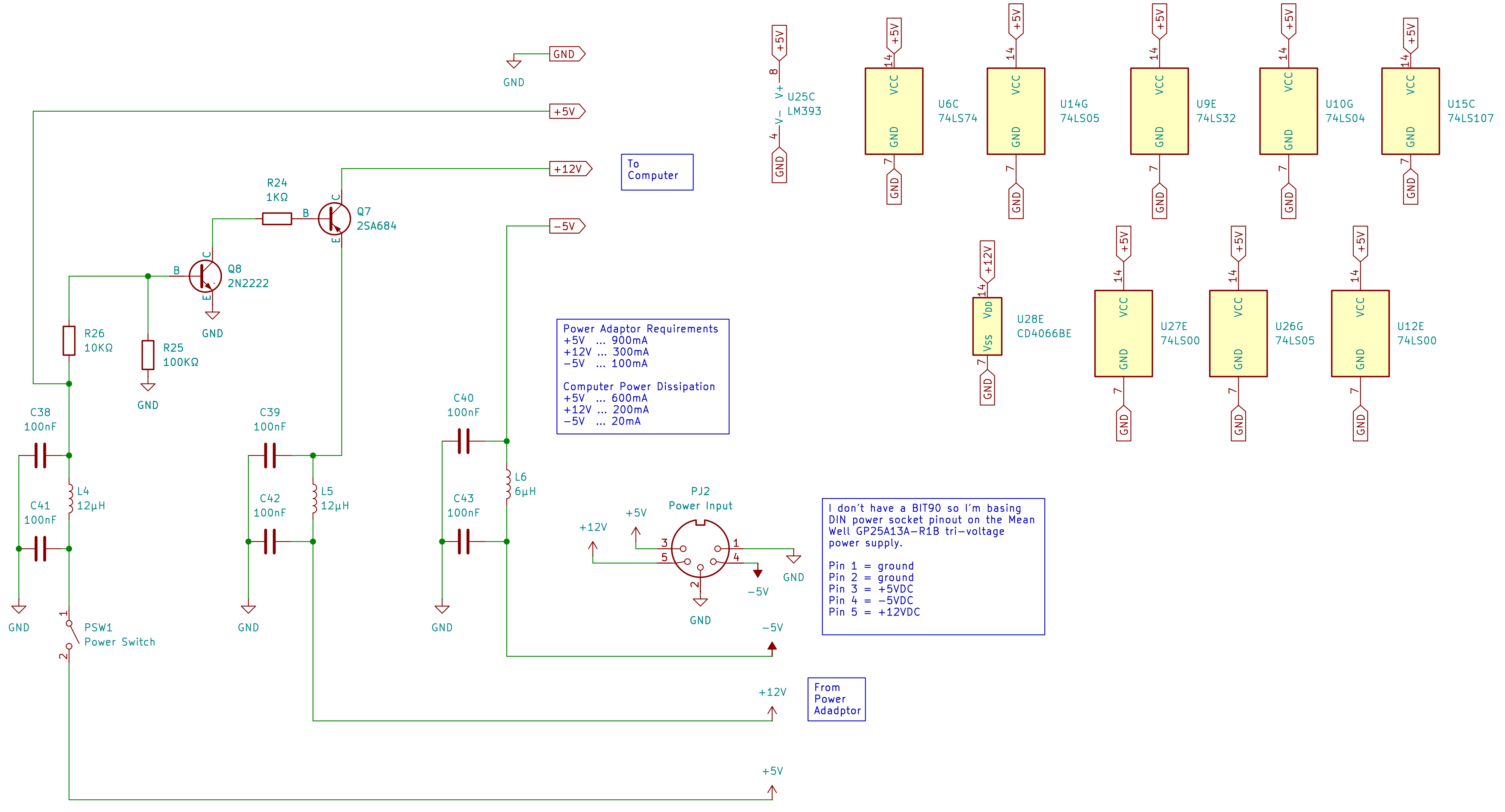
Based on original schematics from BIT Corporation (普澤有限公司)
From Sheet 6−4
[RST] is on Processor sheet

Brett Hallen

Sheet: /Keyboard/
File: BIT90_Keyboard.kicad_sch

Title: BIT90 Keyboard

Size: A3	Date: 5/JUN/2026	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 2/9



Based on original schematics from BIT Corporation (普澤有限公司)
From Sheet 6-4

Brett Hallen

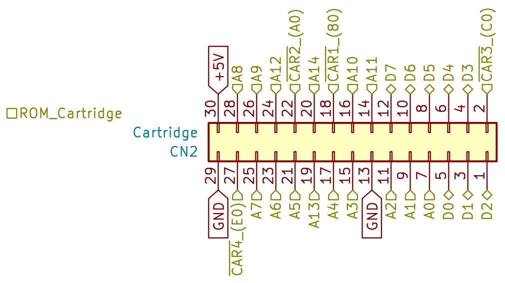
Sheet: /Power/

File: BIT90_Power.kicad_sch

Title: *BIT90 Power*

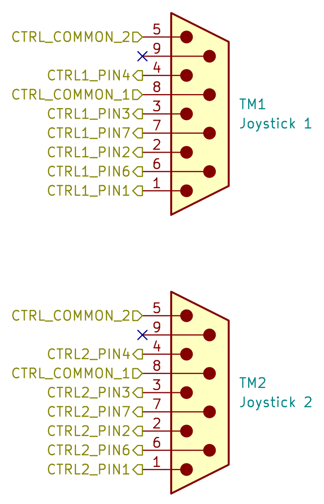
Size: A3 | Date: 5/JUN/2026 | **Rev: A**

KiCad E.D.A. 9.0.9 | Id: 3/9

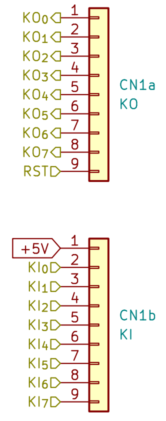


Joysticks

Doesn't match ColecoVision as far as I can tell



Keyboard Connector



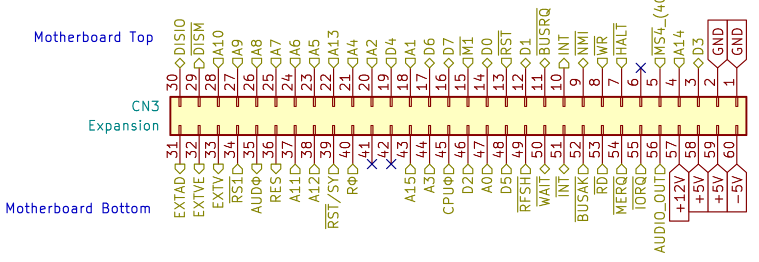
Original keyboard connector CN1 is two 9-pin Molex male connectors, pins numbered 1-18

I am guessing at the connection between keyboard matrix (KI & KO) and actual connector pin ...

Not relevant for recreation as keyboard will be intergratd to motherboard.

System Expansion (Edge Connector)

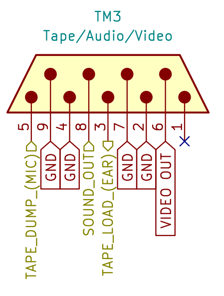
Pin 6 = EN_20 on CV
On BIT90 we have BA2 instead



{Yes yes, the label directions are probably wrong & inconsistent, remind me later to sort out}

AUDΦ is marked as audio clock on CV schematics
No idea how it's generated on the BIT90 as I can't see it specifically noted.

Combination Tape/Audio/Video



Video Output: 75Ω load, 0-1V (white)

Audio Output: 1.5Vp-p

Tape Load (EAR): 500mVp-p
Tape Dump (MIC): 90mVp-p

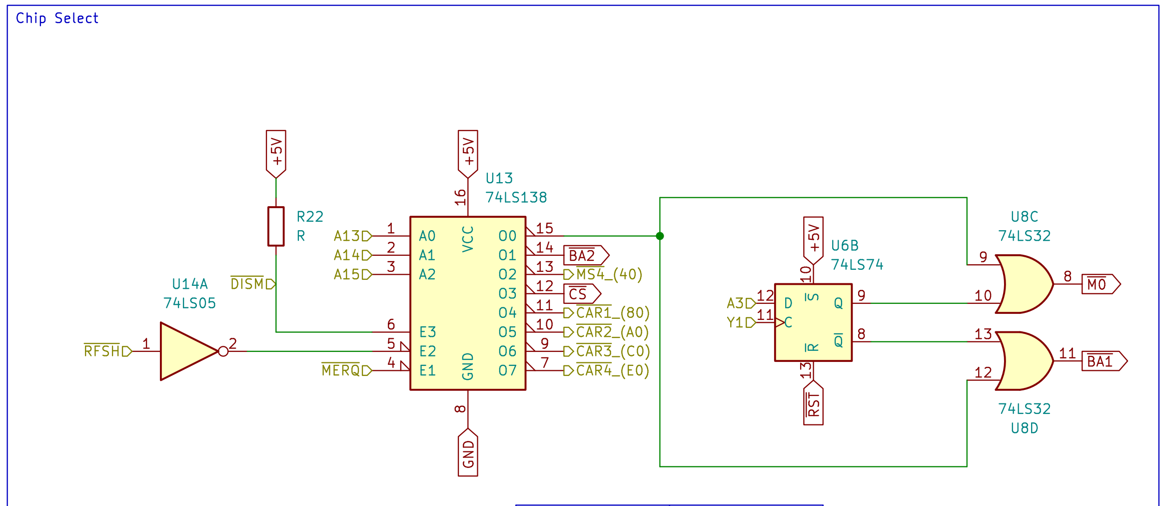
Based on original schematics from BIT Corporation (普澤有限公司)

Brett Hallen

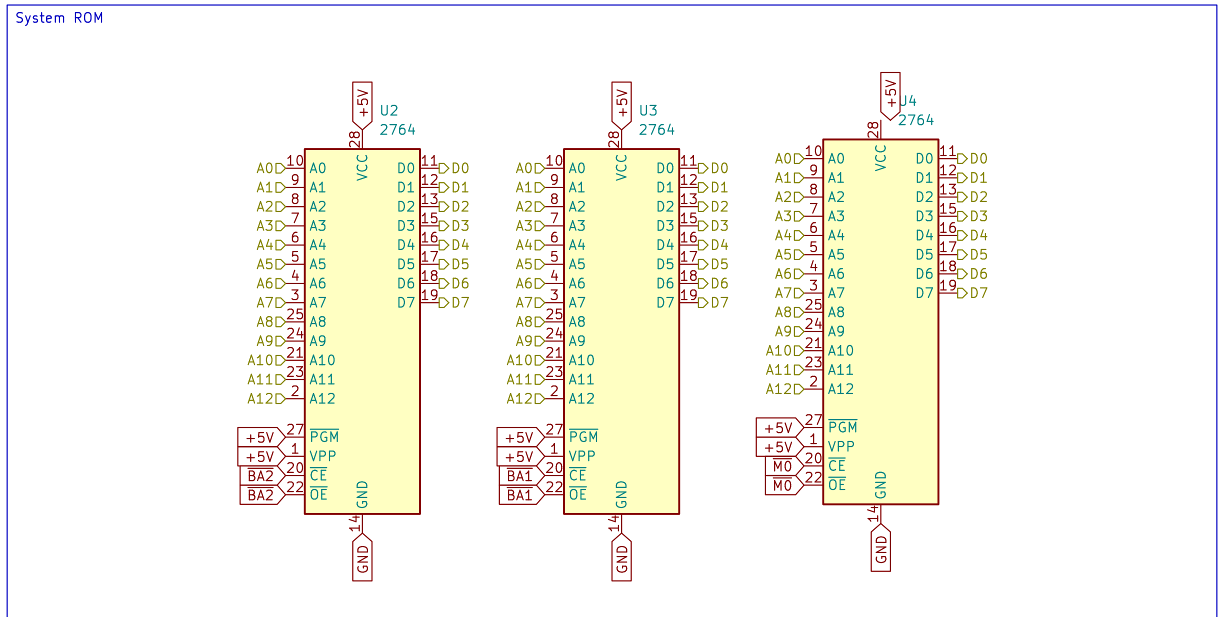
Sheet: /Interfaces/
File: BIT90_Interfaces.kicad_sch

Title: BIT90 Interfaces

Size: A3 | Date: 5/JUN/2026 | Rev: A
KiCad E.D.A. 9.0.9 | Id: 4/9

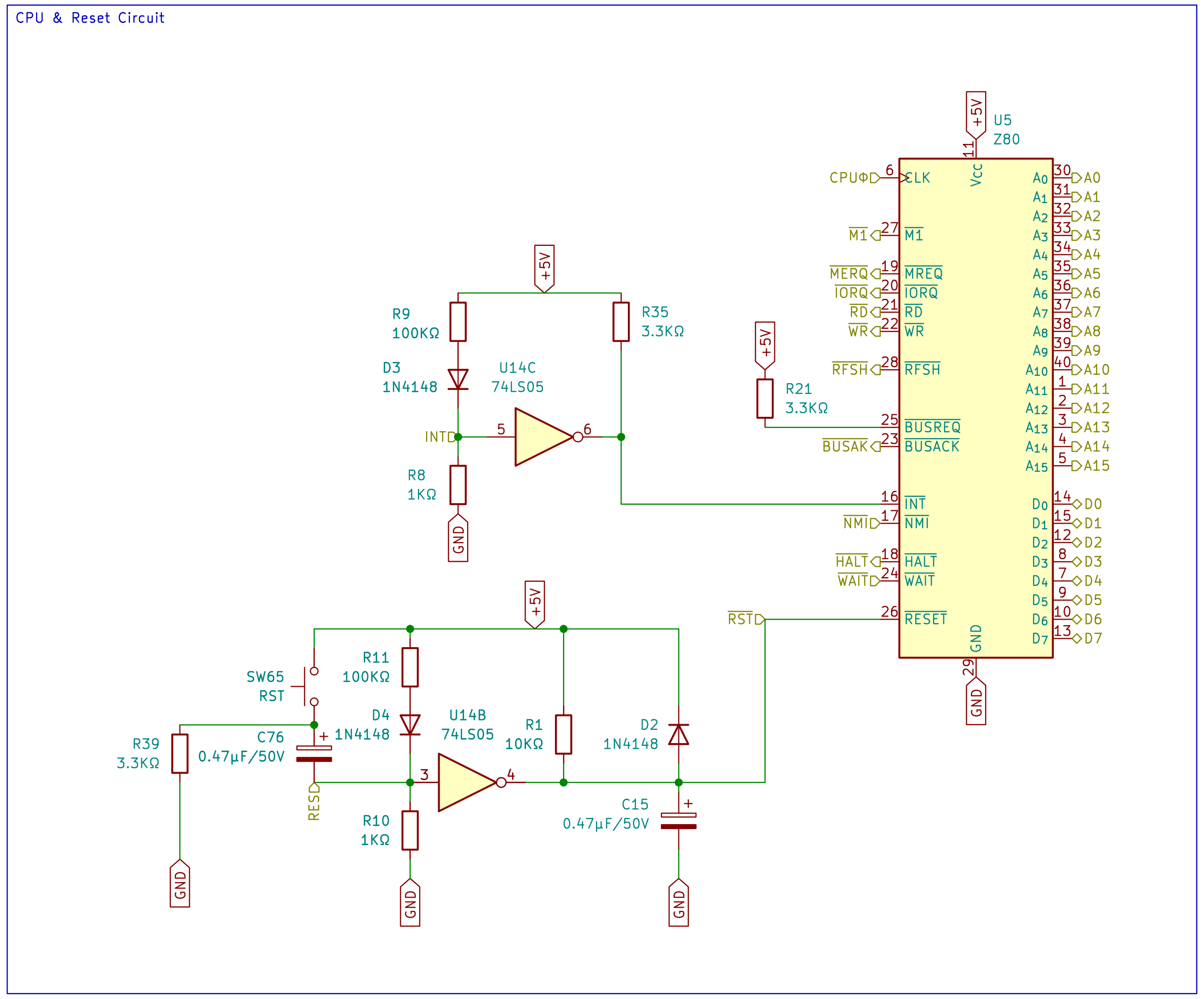


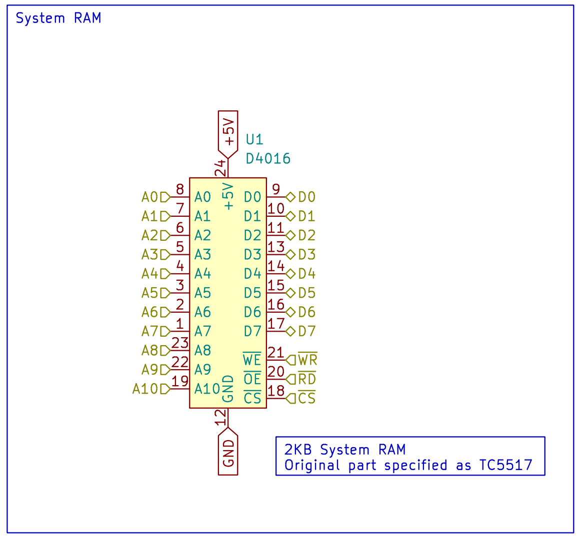
$\overline{M0}$	Monitor
$\overline{BA1}$ & $\overline{BA2}$	BASIC
$\overline{CS}$	System RAM
$\overline{CAR1}$ to $\overline{CAR4}$	Cartridge



Note! Original board can have 2364 masked ROMs (24-pin), 2764 EPROMs (28-pins), or both.
There are three jumper blocks (J1/J2/J3) that configure each ROM socket.
For simplicity they are all represented as 2764 here.

Example:
U2 ... D32522E/MONITOR3 (2364)
U3 ... D32521E (2764)
U4 ... D32351E/BIT-99C1 (2364)



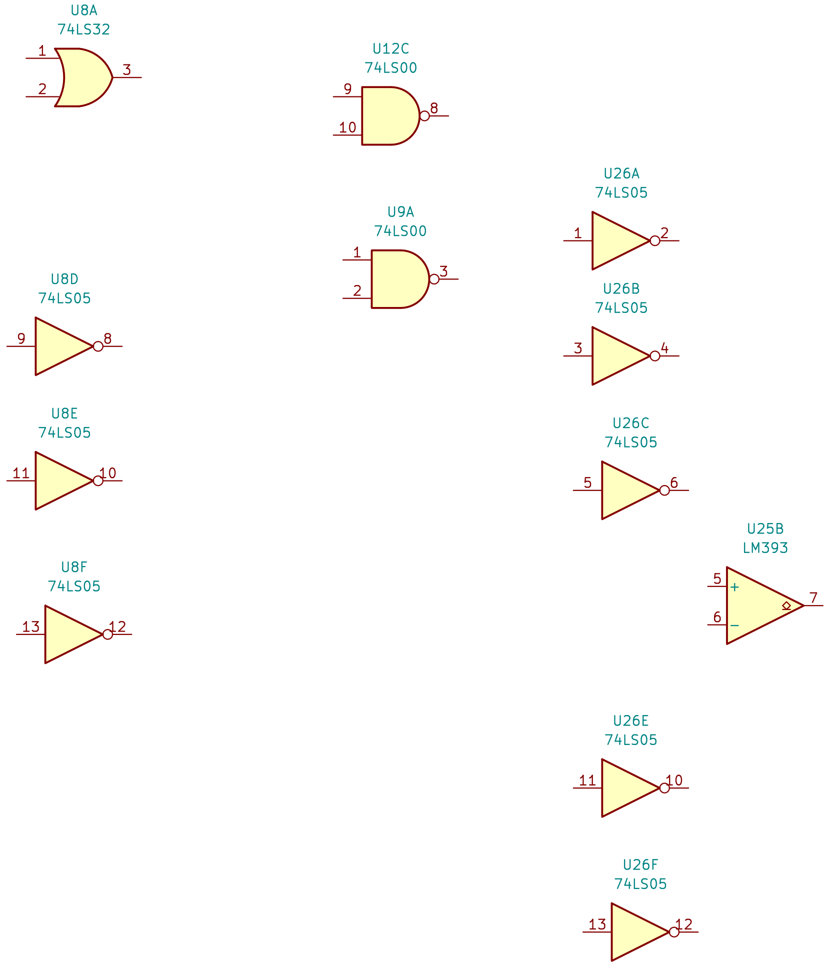


Based on original schematics from BIT Corporation (普澤有限公司)
From Sheet 6-1

Brett Hallen
Sheet: /Processor/
File: BIT90_Processor.kicad_sch

Title: BIT90 Processor

Size: A3 Date: 5/JUN/2026 **Rev: A**
KiCad E.D.A. 9.0.9 Id: 5/9



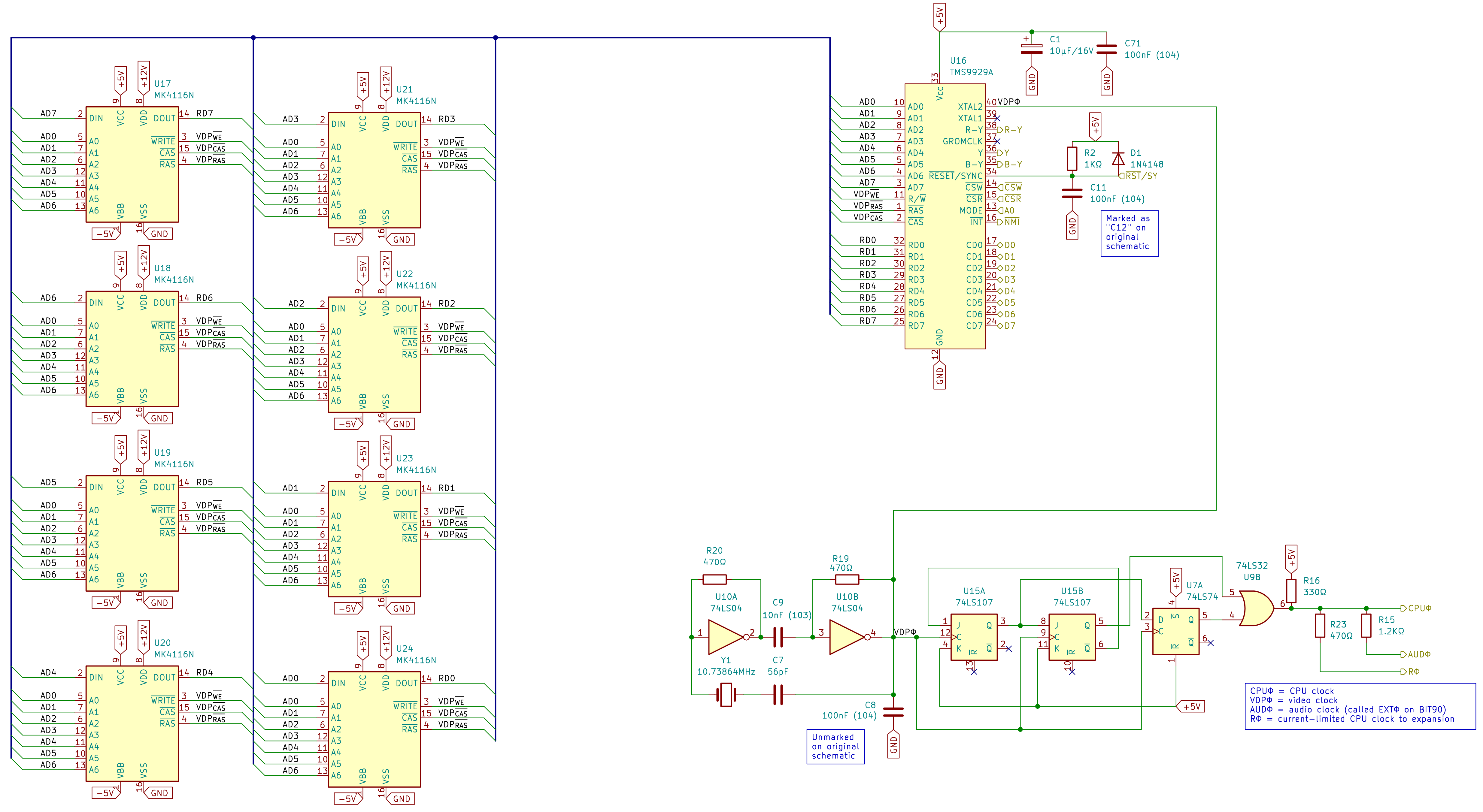
Based on original schematics from BIT Corporation (普澤有限公司)

Brett Hallen

Sheet: /Spare Gates/
File: BIT90_Spares.kicad_sch

Title:

Size: A3	Date: 5/JUN/2026	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 6/9



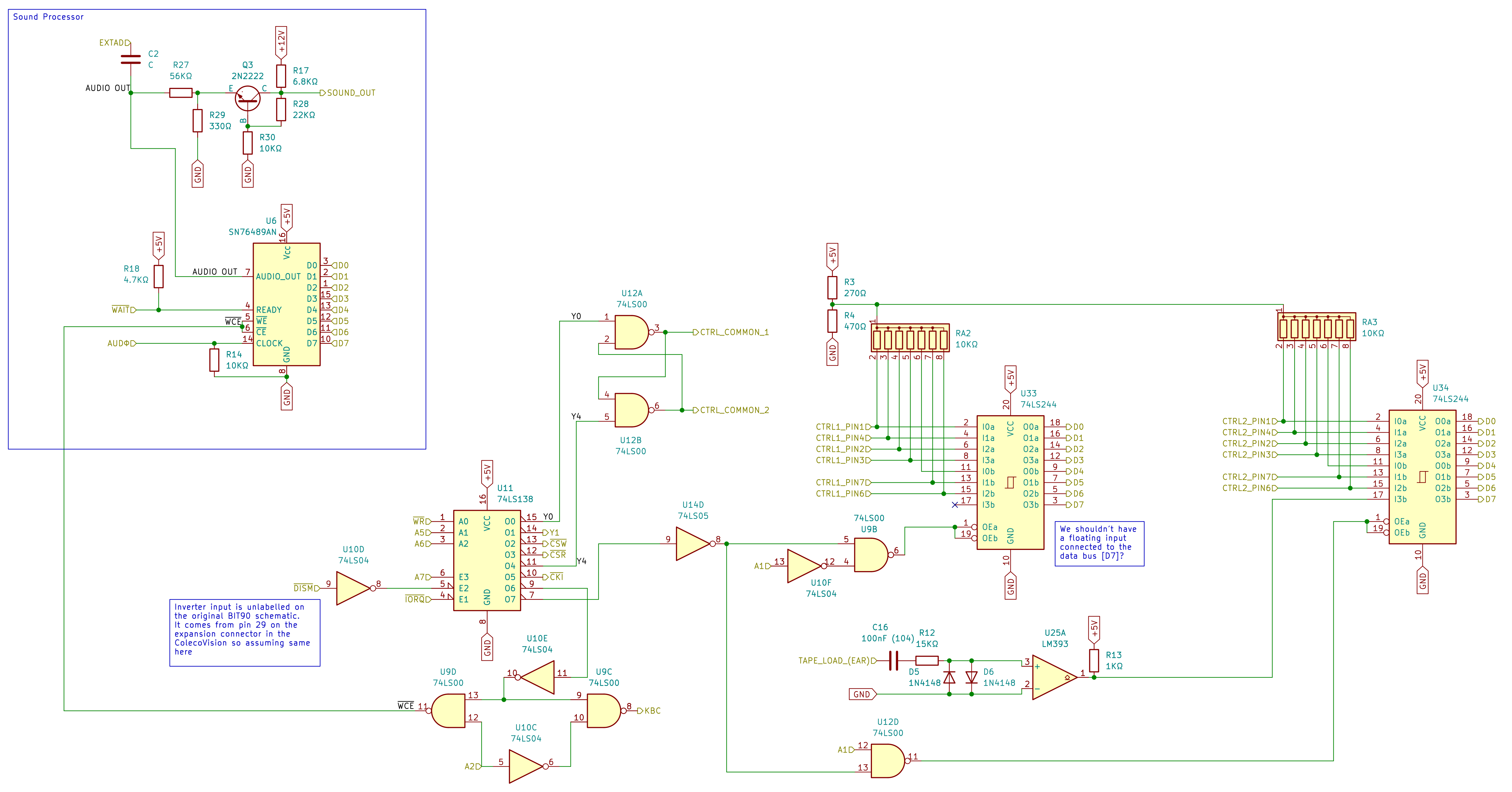
Based on original schematics from BIT Corporation (普澤有限公司)
From Sheet 6-2

Brett Hallen

Sheet: /Video Control/
File: BIT90_Video_Control.kicad_sch

Title: BIT90 Video Control (PAL)

Size: A3 | Date: 5/JUN/2026 | Rev: A
KiCad E.D.A. 9.0.9 | Id: 7/9



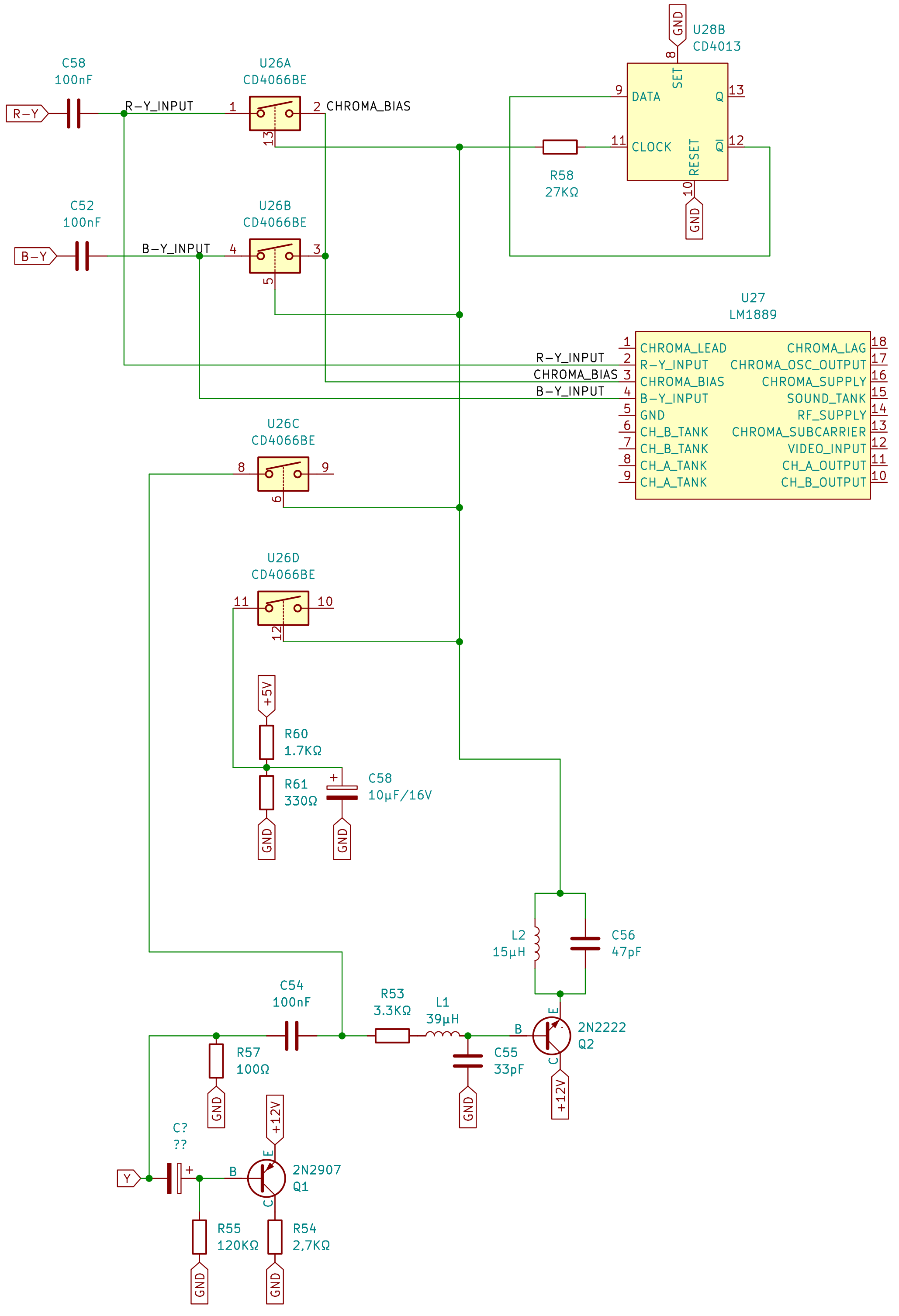
Based on original schematics from BIT Corporation (普澤有限公司)
From Sheet 6-3

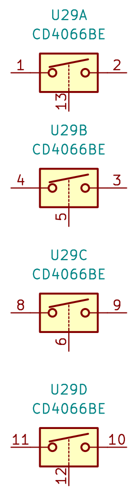
Brett Hallen

Sheet: /Joystick, Audio, Tape Out/
File: BIT90_Joystick_Audio_TapeOut.kicad_sch

Title: BIT90 Joystick, Audio, Tape Out

Size: A3	Date: 5/JUN/2026	Rev: A
KiCad E.D.A. 9.0.9		Id: 8/9





Pffft, is there any point in completing this ... I'm not going to use it in the new design ...

Moved the audio/sound out to Joystick/Audio/Tape Out schematic sheet

Based on original schematics from BIT Corporation (普澤有限公司)
From Sheet 6-5

Brett Hallen
Sheet: /Video Encoder (PAL)/
File: BIT90_Video_Encoder_PAL.kicad_sch

Title: *BIT90 Video Encoder (PAL)*

Size: A3 | Date: 5/JUN/2026 | **Rev: A**
KiCad E.D.A. 9.0.9 | Id: 9/9